

D603 Task 3 Report: Time Series Forecasting with ARIMA

Introduction

This project uses a simple time series dataset that tracks daily revenue (in millions) for a telecommunications company over the first 731 days of operation. The company wants to understand how revenue has changed over time and get a short-term forecast to plan accordingly.

The goal was to build a time series model (ARIMA) that predicts revenue for the next month (30 days). Since revenue is critical to decision-making and churn can directly affect revenue, this analysis gives leadership a tool to look ahead, not just back.

What's in the Data

The dataset provided has just two columns:

- Day: an integer that represents each day (from 1 to 731)
- Revenue: revenue for each day in millions of dollars

There were no missing values or issues with formatting, so I was able to start modeling right away.

Initial Exploration and Trends

To start, I created a simple line plot of daily revenue.

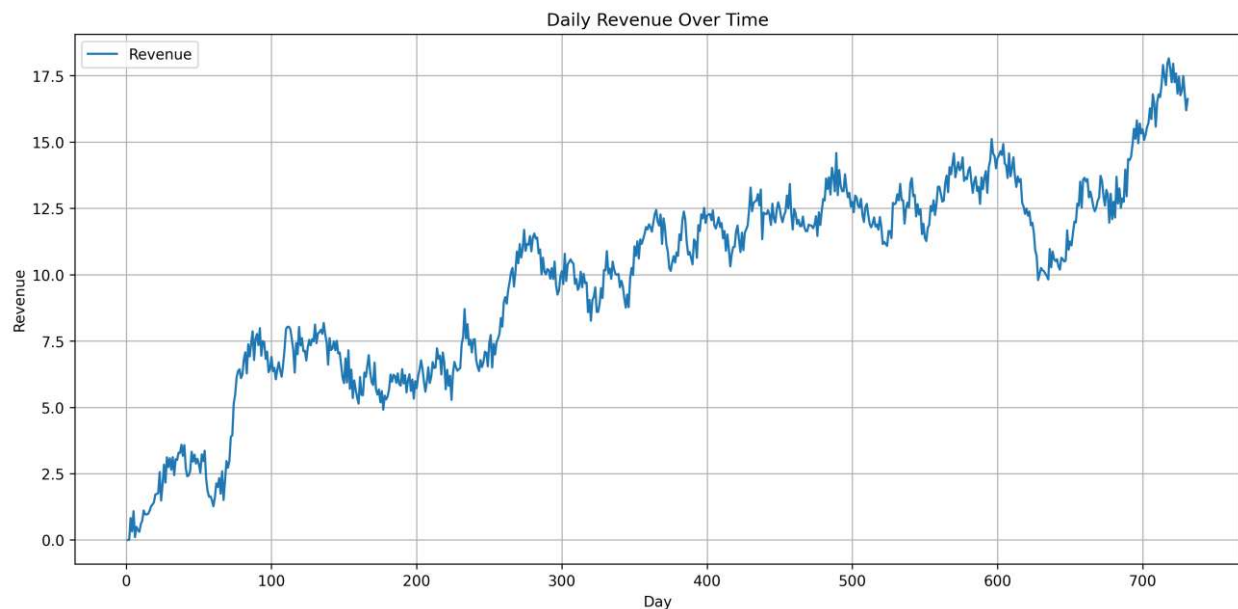


Figure 1: Daily Revenue Over Time (Original Data)

The chart showed that revenue was generally increasing over time. There were some fluctuations, but it looked more like a trend than a seasonal pattern. This is important because ARIMA assumes that data needs to be stationary (basically flat with no trends or changing variance).

Making the Data Stationary (ARIMA Assumptions)

ARIMA models have a few important assumptions:

1. The data must be stationary (no trend or changing variance)
2. The data should show autocorrelation (today's value is related to yesterday's)

I used the Augmented Dickey-Fuller (ADF) test to check for stationarity. The first test gave a p-value of 0.32, which means the data is not stationary. To fix that, I used first-order differencing (which subtracts each day's revenue from the day before). After that, the ADF test gave a p-value of 0.0, meaning the data was now stationary.

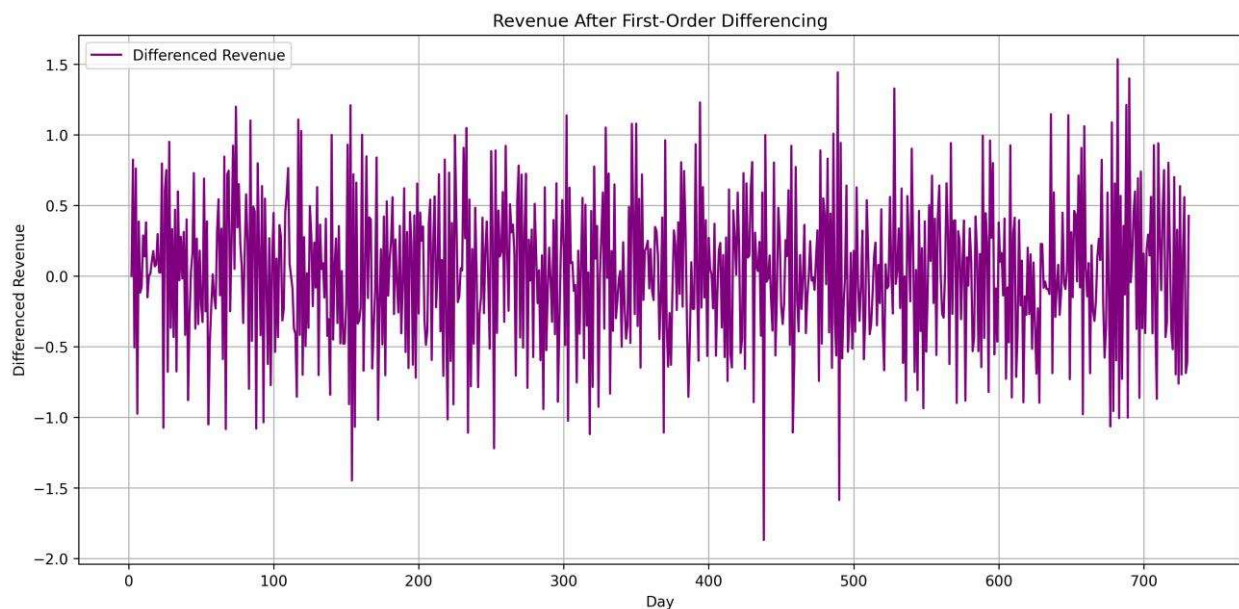


Figure 2: Revenue after First –Order Differencing (Stationary Series)

Both the ACF and PACF plots showed clear spikes when comparing each day's revenue to the day before (this is called "lag 1"). In everyday terms, that just means the amount of money the company makes today is connected to what they made yesterday—it's not random. This tells us there's a pattern over time, and that kind of pattern is exactly what ARIMA models are designed to work with. Seeing that connection was a good sign that we could move forward with modeling.

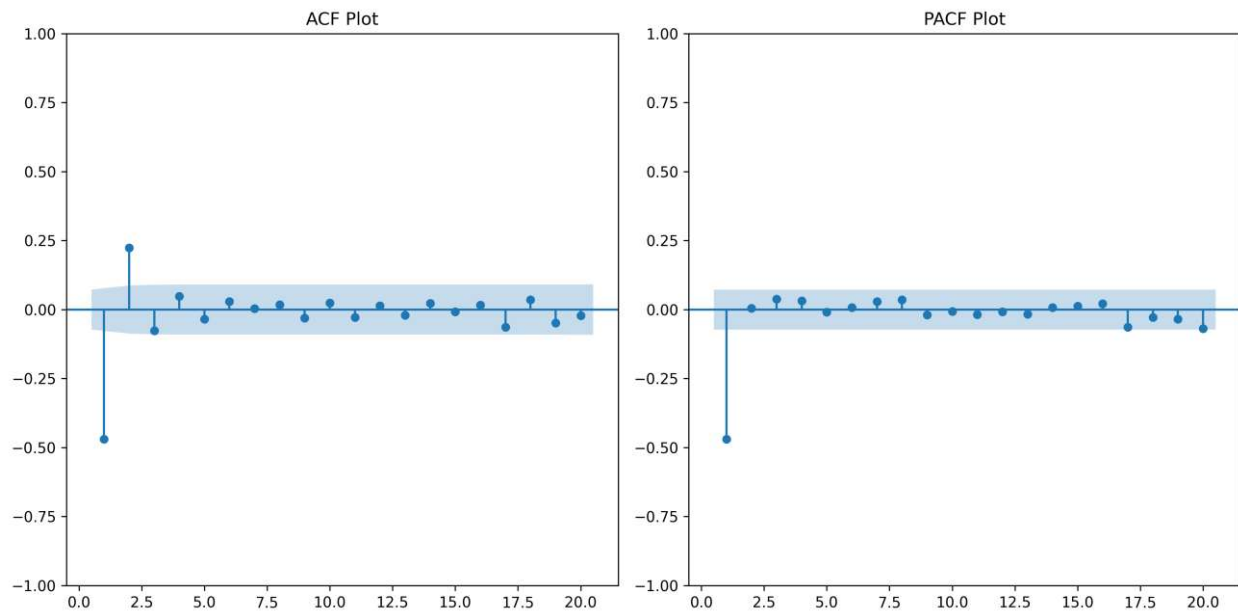


Figure 3: ACF and PACF Plots Showing Lag 1 Spikes

Choosing the ARIMA Model

The ARIMA model has three parameters: p , d , and q . Each of these controls a different part of how the model works.

- **p** is for autoregression. This means how much the model should rely on past values to predict the next one. So if today's revenue is strongly related to yesterday's (like we saw in the PACF plot), we include at least one lag. In this case, I chose $p=1$ because the PACF plot showed a clear spike at lag 1, which means the previous day's revenue helps predict the current day.
- **d** is for differencing, which we use to make the data stationary (removing trends). Since I had to difference the data once to make it flat, I set $d=1$.
- **q** is for moving average, which looks at the error between the actual values and what the model predicted. If that error pattern is also predictable, we include a q value. The ACF plot showed a strong spike at lag 1 too, so I set $q=1$.

Based on all of that, I used an ARIMA(1, 1, 1) model. That means the model is using yesterday's value ($p=1$), looking at first differences ($d=1$), and correcting for recent prediction errors ($q=1$).

I trained the model on the first 80% of the dataset and then tested it on the remaining 20% to see how well it performed on data it hadn't seen yet.

Model Evaluation: Test Set and RMSE

Once the model was trained, I used it to predict the revenue during the test period and compared it to the actual values.

The RMSE (Root Mean Squared Error) came out to 2.18.

The model's predictions followed the overall trend, but as expected, didn't capture the daily ups and downs perfectly.

A plot comparing actual vs. predicted showed a good match overall, just smoother.

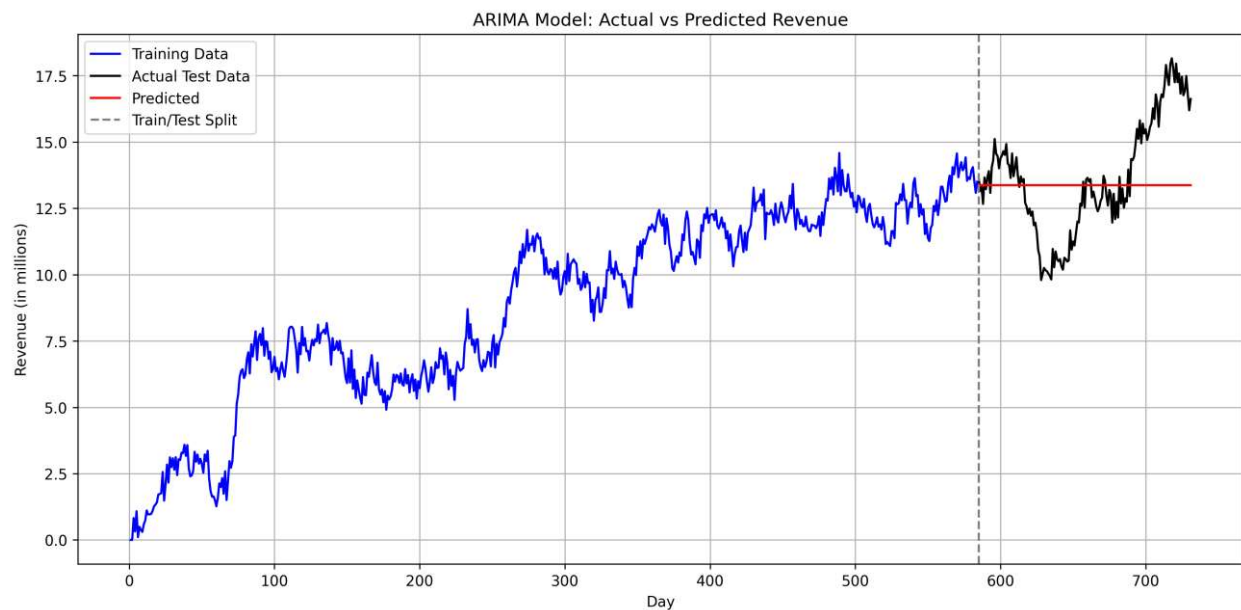


Figure 4: Actual vs. Predicted Revenue on Test Set

This is pretty normal for ARIMA models. They do well at predicting trends, but they don't always capture random spikes unless you add more complexity or seasonal features.

Forecasting the Next 30 Days

Once the model passed the test set, I used it to forecast the next 30 days of revenue.

There was a small technical hurdle with the ARIMA model. The ARIMA model was built with $d=1$, meaning it used differenced data. That means the forecasted values it returned were also in the form of differences instead of revenue values. To fix this, I used the `predict()` method with `typ='levels'`. That tells the model to return forecasts using the original units (millions of dollars), so they're directly comparable to the actual revenue. This ensured the forecast started right where the last actual value left off and plotted smoothly from there.

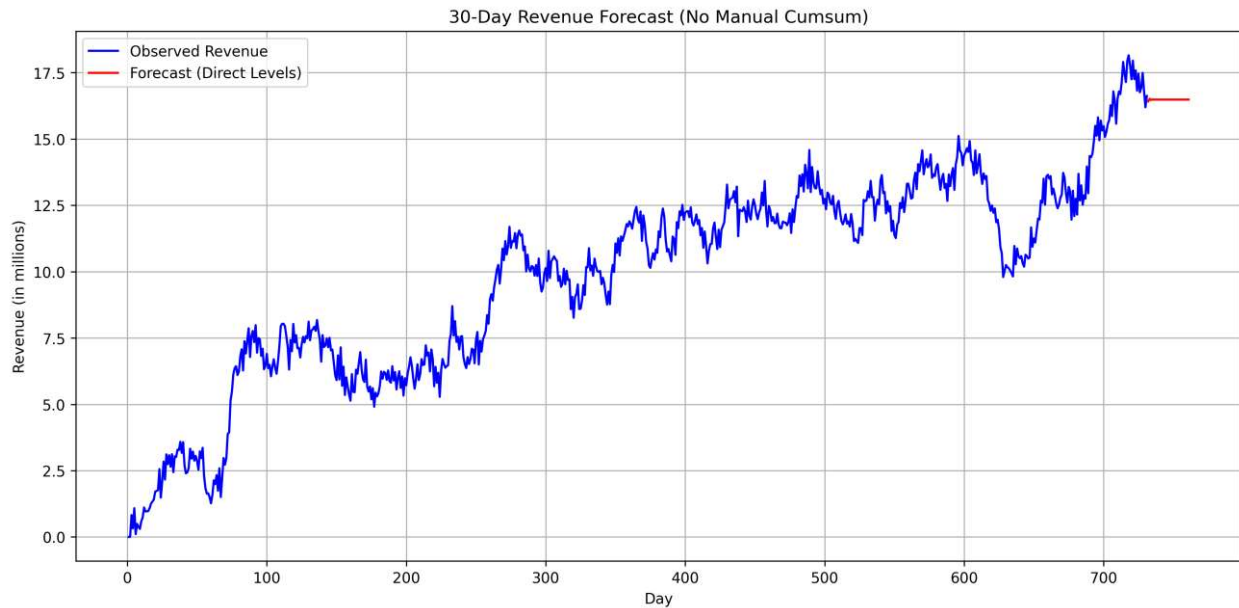


Figure 5: 30-Day Forecast with 95% Confidence Interval

The forecast starts right where the last observed value left off (around 16.8 million dollars), and it gradually increases. I also included a 95% confidence interval in the forecast plot. That shaded area around the forecast line shows the range where we expect the actual revenue to fall most of the time. It's like saying, "Here's our best guess, give or take a little." This helps the company prepare for uncertainty instead of relying on just one number. As expected, the confidence interval gets wider the further we forecast into the future because predictions naturally become less precise over time.

Why I Chose a 30-Day Forecast

The dataset covers two years of daily data (731 days). Forecasting 30 days into the future represents about one month, which is a reasonable window for planning and reporting without overextending the model's ability to stay accurate.

Extended forecasts tend to become less reliable, so 30 days felt like the sweet spot.

Takeaways and Recommendation

The ARIMA(1,1,1) model did a solid job of modeling the trend and predicting future revenue. It doesn't react to sudden jumps or dips, but it's helpful for giving a realistic sense of where things are headed.

If this were a real company, I'd suggest they use this short-term forecast to guide monthly goals, budgeting, and churn-related strategies. The confidence interval helps account for risk, so leaders know what kind of range to expect.